Exploring Total Cost of Tier 1 Trauma Center Stays in New York State

Lucas Henry (25252089) - Spring 2026 - Module: Predictive modeling

link to github

Data

Where did it come from?
What does it look like?

Individual level, **population** discharge data from 2024 New York State hospitals

Link to data source

- De-identified
- Individual Level
- Treatment/outcomes(next slide)

What's in this Dataset?

Rows	Columns
2.2M	33

Columns (33)

Column Name	Description	API Field Name	Data Type
Health Service Area	A description of the Health Service Area (HSA) in which the hospital is located.	health_service_area	Text
Hospital County	A description of the county in which the hospital is located.	hospital_county	Text
Operating Certificate Number	The facility Operating Certificate Number assigned by NYS Department of Health.	operating_certificate_number	Text
Permanent Facility Id	Permanent Facility Identifier.	permanent_facility_id	Text
Facility Name	The name of the facility where services were performed based on the Permanent Facility Identifier (PFI), as maintained by the NYSDOH Division of Health Facility...	facility_name	Text

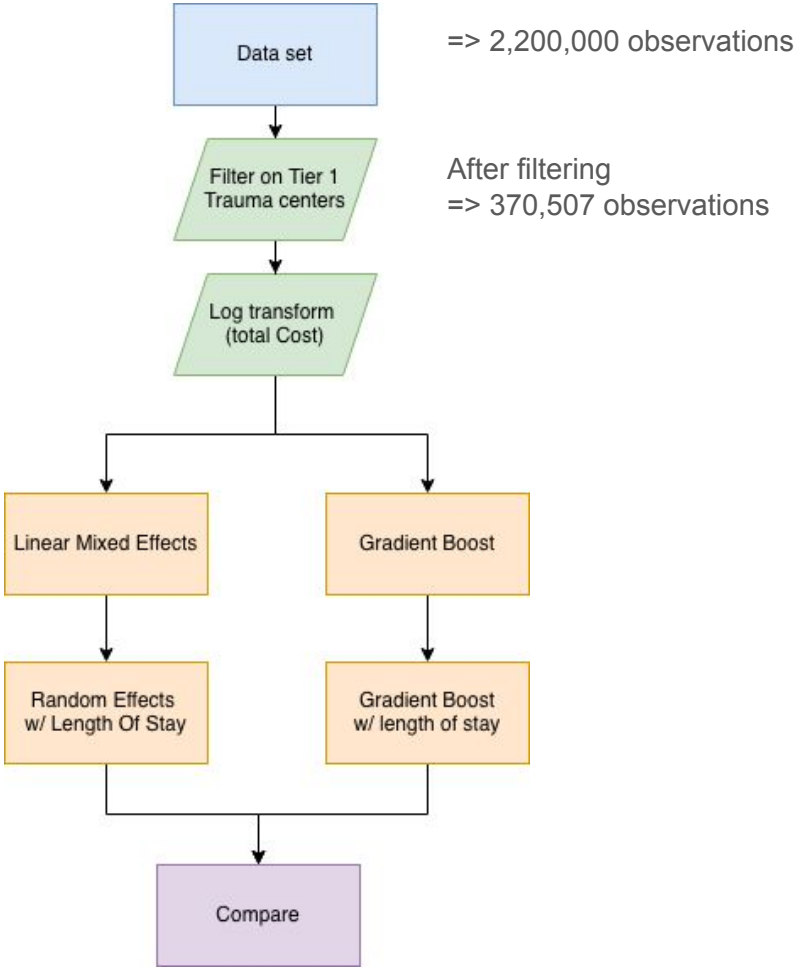
Selected variables

Response/Predictors

Column name	Description	Example
Total Costs	Estimated cost for the discharge	1233,222,333,1234,....
Disposition	Patient destination or status	Expired, home, assisted living,...
APR.MDC.Description	General condition description	DISEASES AND DISORDERS OF THE CIRCULATORY SYSTEM,...
Facility.Name	Facility Name	ERIE COUNTY MEDICAL CENTER, NYU...
Age.Group	Age Group	0-17, … , 70 or Older
Gender	Gender	Male, Female, Unknown
APR.Severity.of.Illness.Code	Severity of Illness Code	4 3 2 1
Payment.Typology.1	Payment typology (how patient pays)	Medicaid, Medicare, Private, Self pay, …
Type.of.Admission	Type of admission	Emergency, Trauma, Elective, Urgent, …
Length Of Stay	(Discharge Date - Admission Date) + 1.	54,33,213,120+

Workflow/filtering

- Filter on tier 1 trauma center EDs
- log transform cost
- Fitting 4 models
- All models handle categorical predictors automatically
- Linear mixed effects models with and without length of stay
- Gradient boost with and without length of stay



Cost Summary

- Using costs, not charges
- Costs show actual cost
- Less variable between hospital - federally regulated

Cost Summary (response)

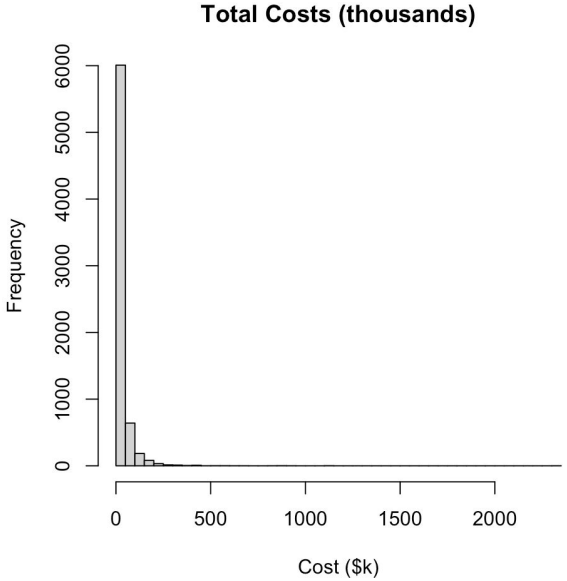
- Right skew (large mean)
- Q1 is just above $9,000
- Q3 is just around $32,000
- Median cost is over $16,000
- Maximum
    - Cost associated with providing care for newborn: $6,000,000
    - Charged: $17,000,000
    - Stay in hospital over 120 days

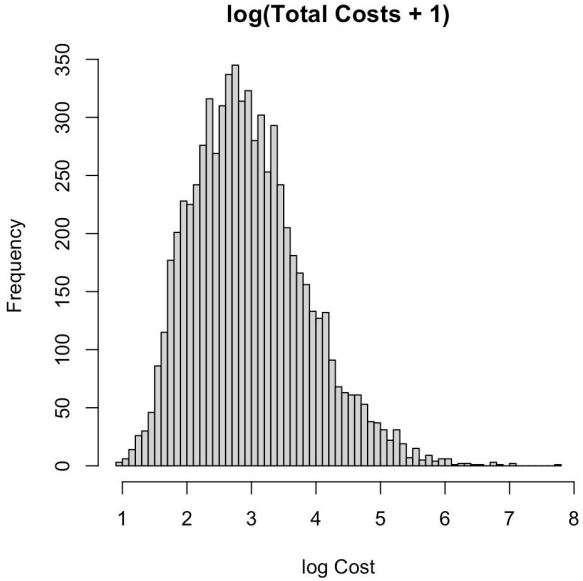
Cost Summary in $1,000s

Statistic	Population	Train	Test
Min	0.04	1.53	1.33
Q1	9.38	9.25	9.58
Median	16.81	16.78	17.27
Mean	30.88	30.77	30.17
Q3	32.25	31.92	32.55
Max	6001.05	2336.32	999.9
SD	59.31	57.62	47.09

Cost Skew

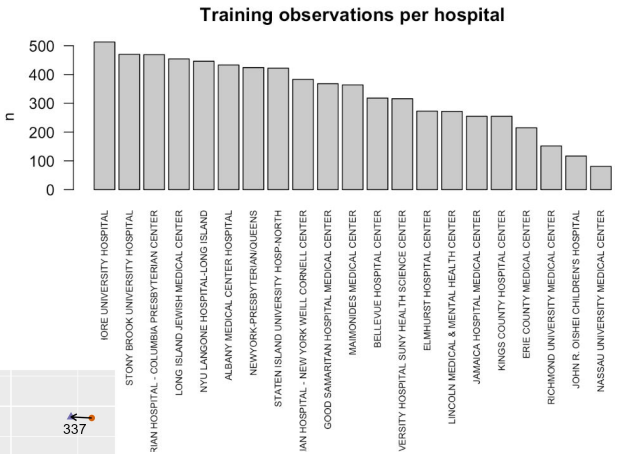
- Log transforming the cost can make it appear more normal, and stabilize variance.
- Linear Mixed effects models assume normal residuals and constant variance

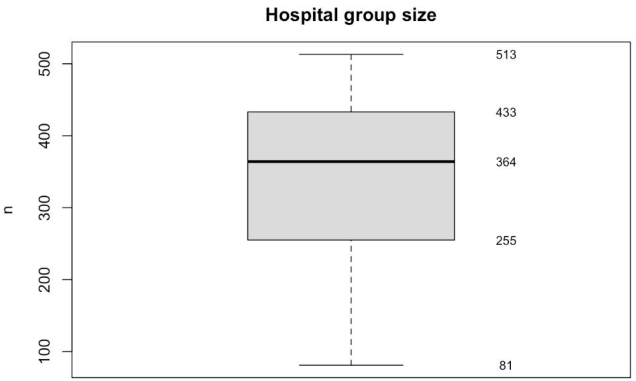


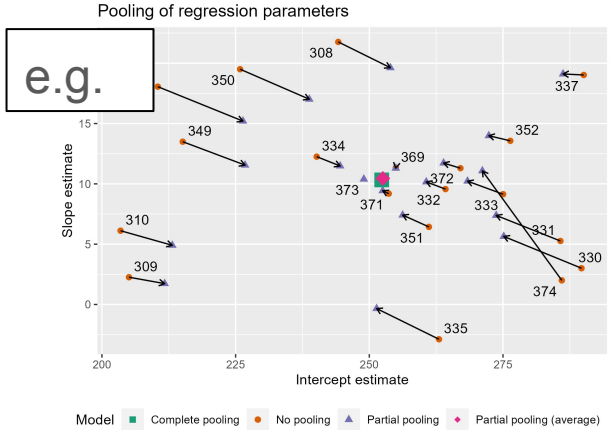


Observations per Hospital

- There is a spread, the hospital with the highest count count is **over 5 times** that of the lowest
- Motivates the random effect in the first model to **pull** low information hospitals **toward grand mean**







Modeling Total Cost Outcome

LME and Gradient Boosting

Mixed Effects Model

with random intercept on tier 1 trauma center

Random effect result (partial pooling)

- Recall pooling
- Hospital intercepts capture 5% of variation
- Already filtering on tier 1 trauma center
- Large residual

```
           grp        var1 var2       vcov     sdcor
Facility.Name (Intercept) <NA> 0.05686434 0.2384625
     Residual        <NA> <NA> 0.44535862 0.6673519
```

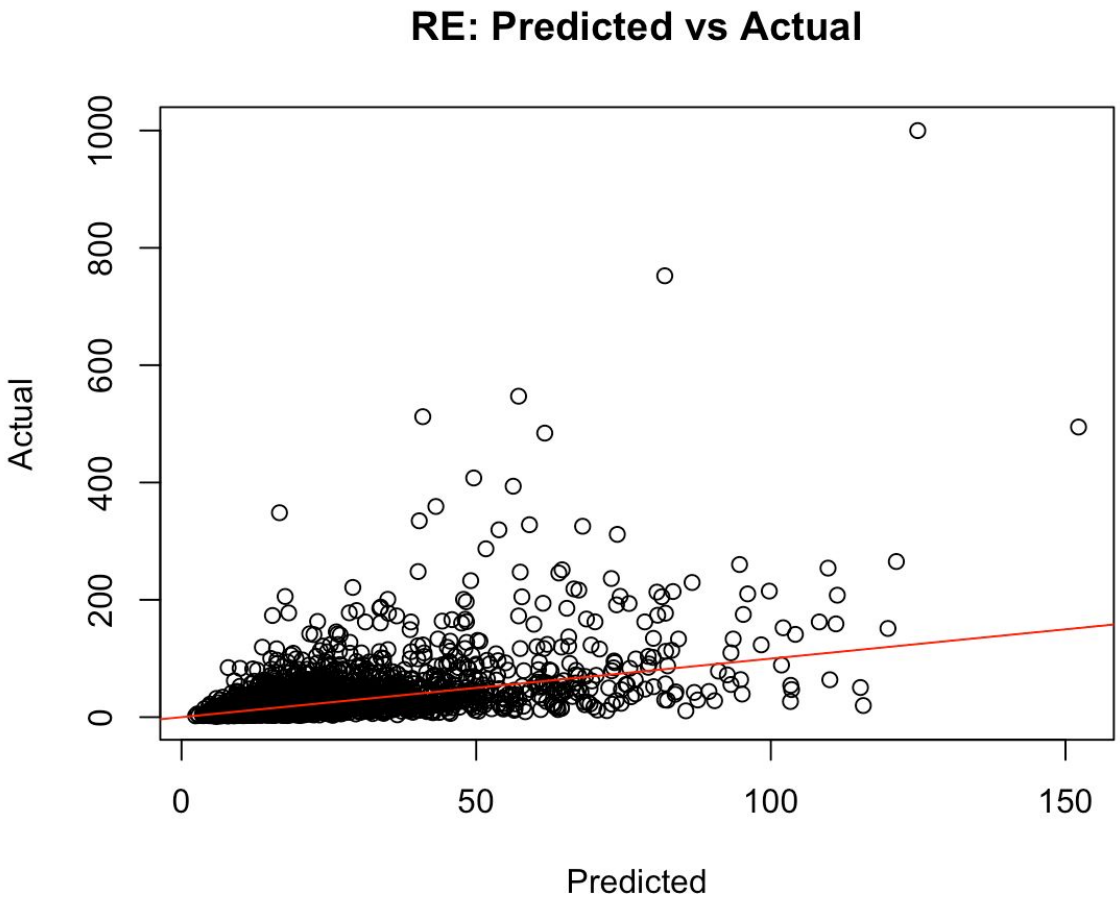
Coefficients

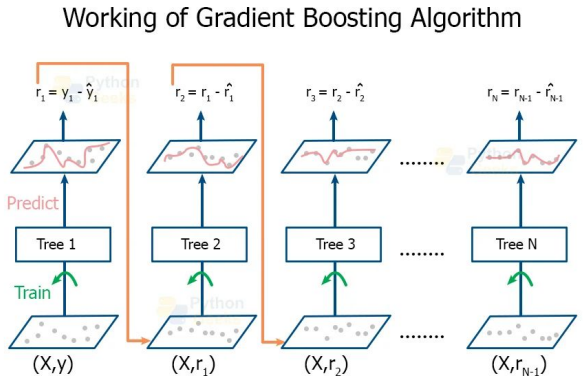
- top 5 coefficients by t statistic

term	Estimate	Std. Error	t value
APR.Severity.of.Illness.Code4	1.37335908	0.03586847	38.2887524
APR.Severity.of.Illness.Code3	0.65006381	0.0255442	25.4485847
APR.MDC.DescriptionMENTAL DISEASES AND DISORDERS	0.69853628	0.0663839	10.522677
APR.Severity.of.Illness.Code2	0.21826624	0.02306928	9.46133615
APR.MDC.DescriptionDISEASES AND DISORDERS OF THE MUSCULOSKELETAL SYSTEM AND CONN TISSUE	0.50492372	0.06216553	8.12224609

Predicted vs Actual Plot

- A well fitting model should hug the diagonal.
- This one is not.
- Will compare models at end
- Weak correlation => uncaptured variation

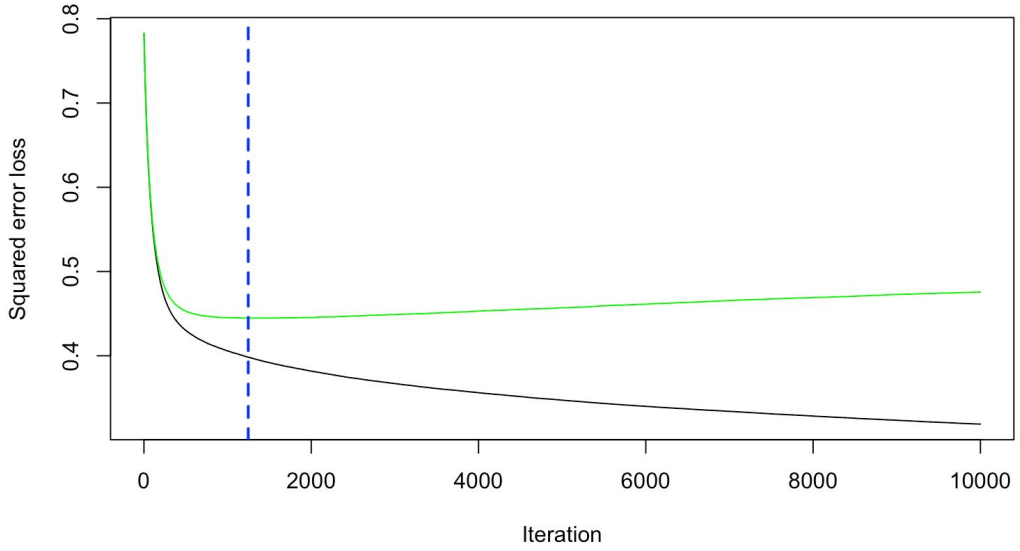




Gradient Boost

Ensemble tree

1. Compute residuals from the current model
2. Fit a new small tree to those residuals
3. Add that tree (scaled by shrinkage) to the model
4. Update residuals, repeat

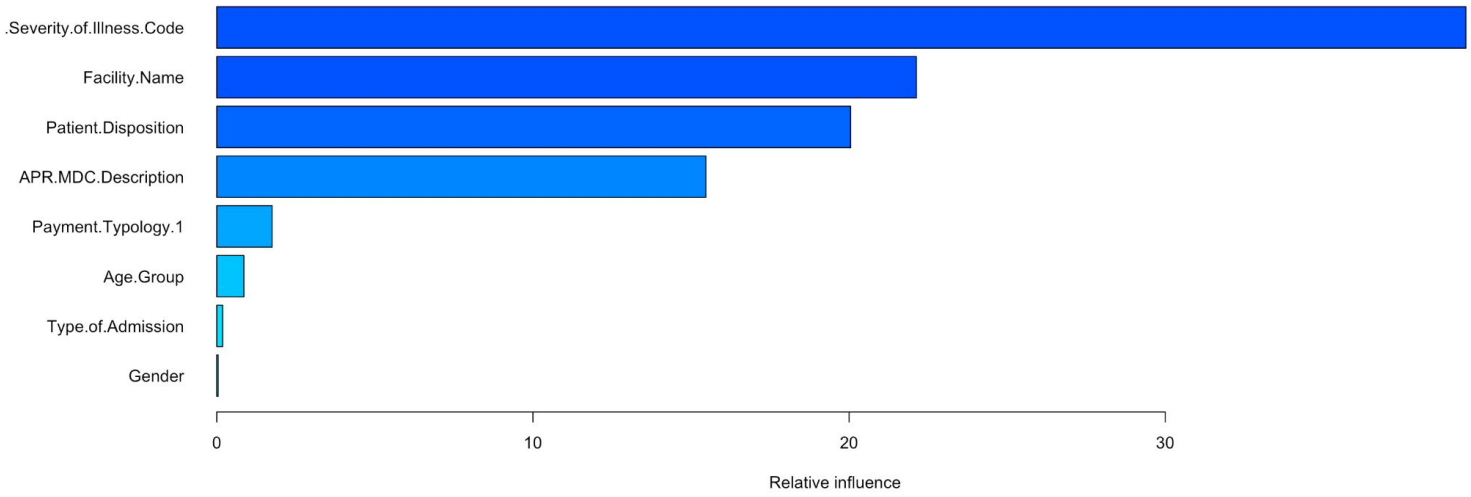


Fitting

- Used cross validation to find the optimal number of trees
  - => 1247
- Using interaction depth 4, shrinkage 0.01

Variable Importance

- Importance shows the variability captured by each predictor

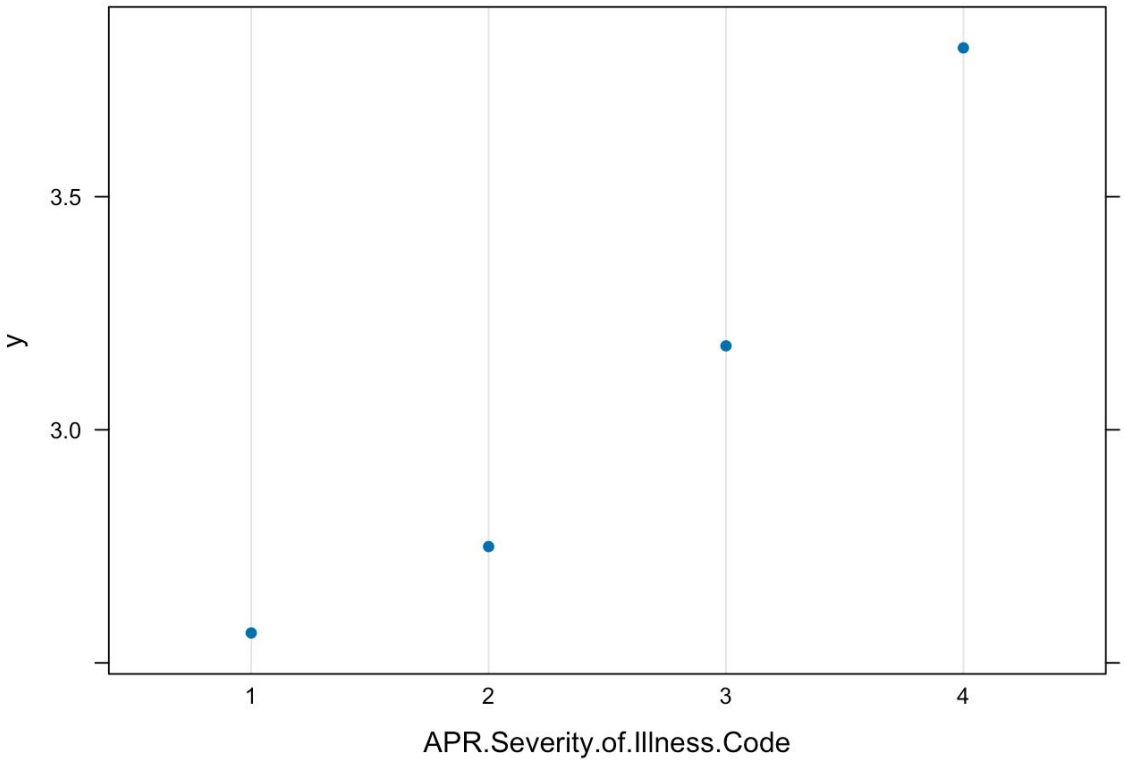


Partial Dependence

- Partial dependence plots show the effect of a single predictor value on total costs after integrating out all other variables

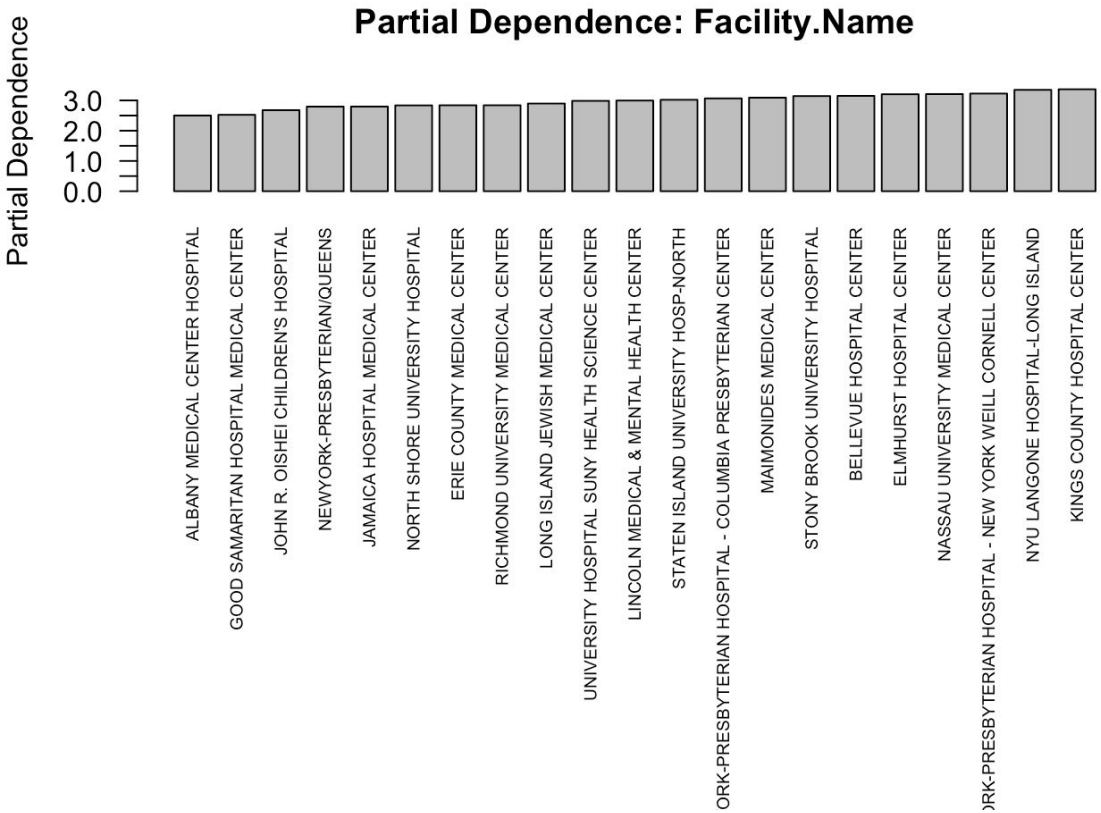
Severity of illness

- Severity of illness has high linear correlation in the response



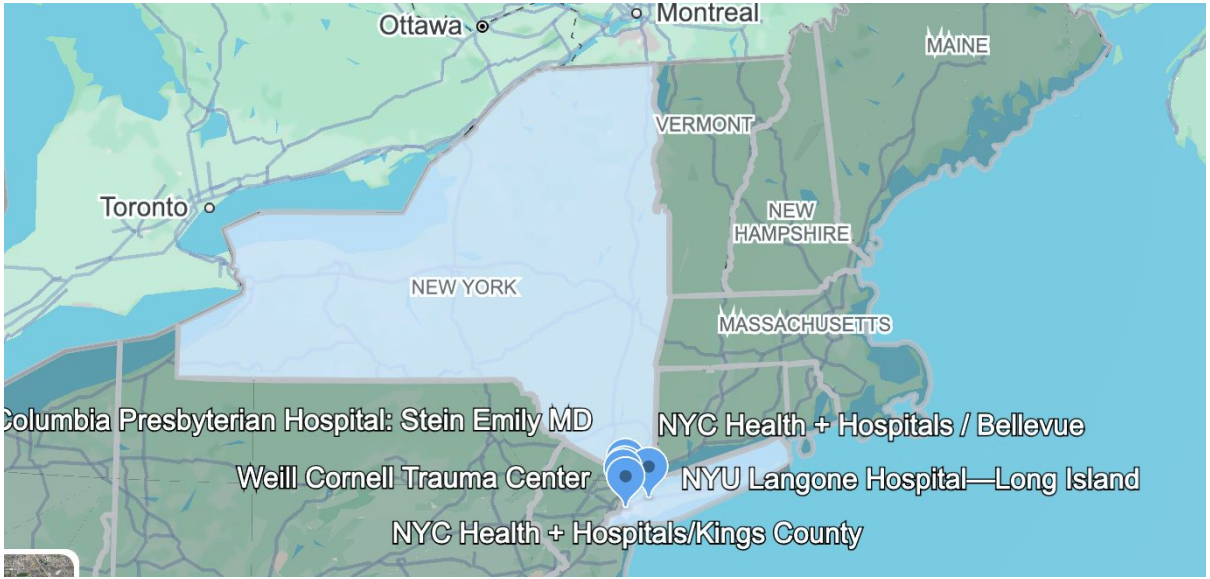
Facility name

- the facility name appears more muted, however it is notable that the top 5 hospitals by importance are in new york city



Hospitals with highest importance on Total Cost

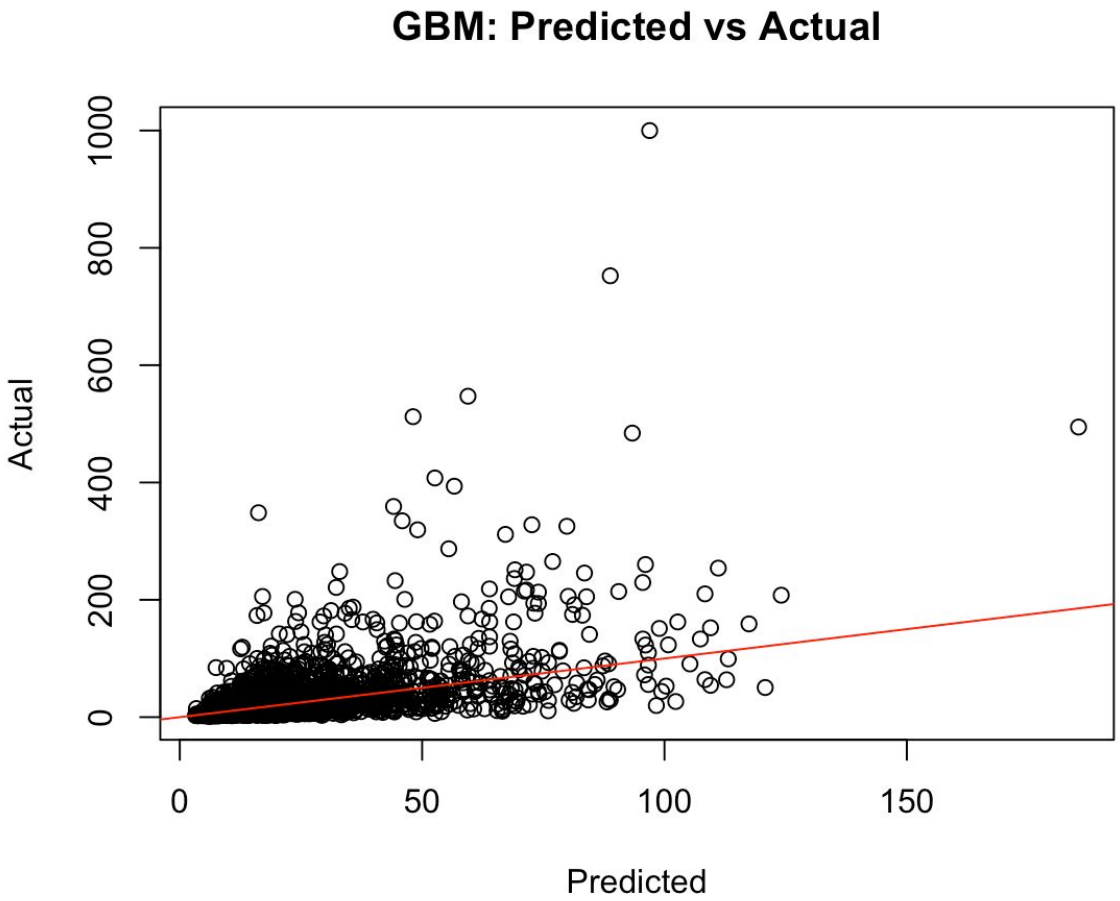
- Interestingly, all in New York City
- Consistent with cost of living/intuition
- Rent is 50-75% less in upstate New York vs New York city*



*https://latestrealestatetrends.com/cost-of-living-upstate-ny/

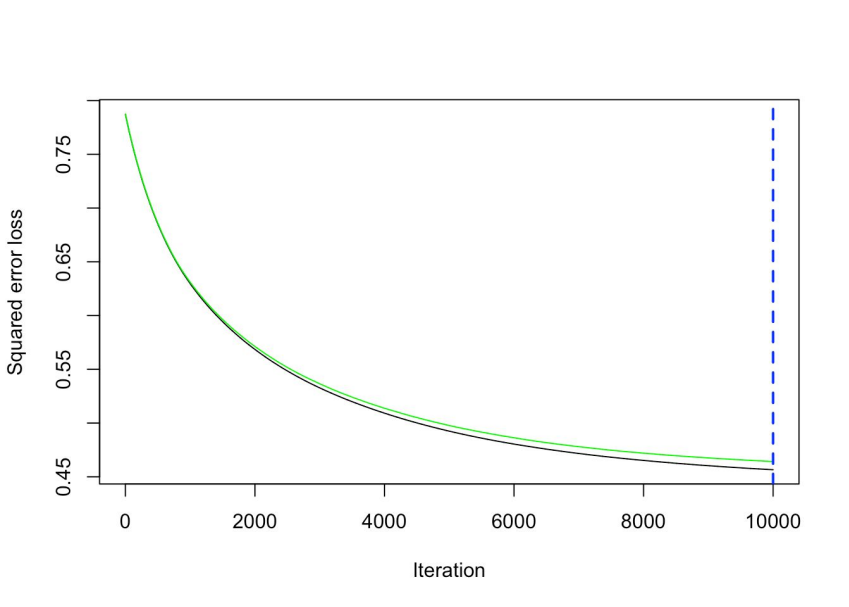
Predicted vs Actual Plot

- A well fitting model should hug the diagonal.
- This one is not really
- Will compare models at end
- Weak correlation => uncaptured variation
- Looks similar to the Linear Mixed Effect model



Other shrinkage (0.001)

- Did not converge in a reasonable iteration number



Add Length of Stay

Idea: Higher correlation with total cost

- fit with same variables

Mixed Effects Model

with random intercept on tier 1 trauma center

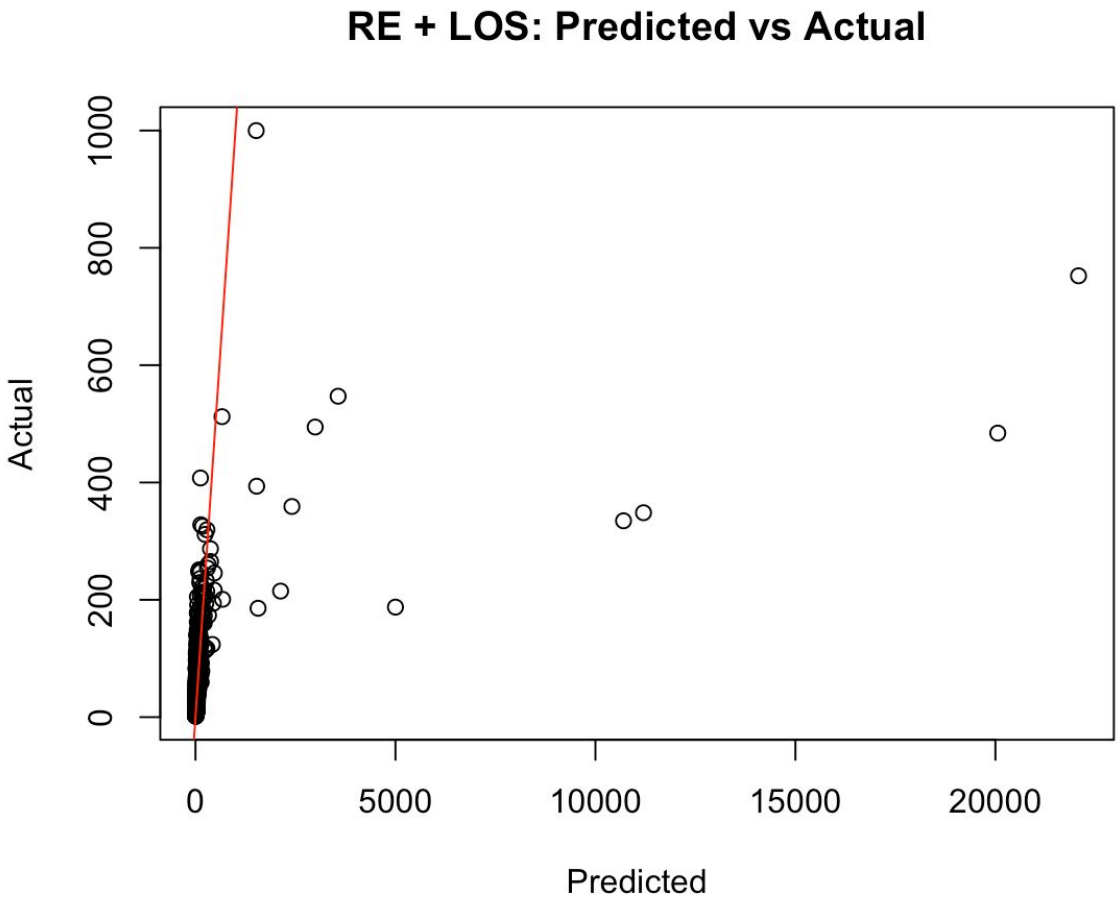
Coefficients

- top 5 coefficients by t statistic

Term	Estimate	Std. Error	t value
Length.of.Stay	0.05661455	0.00073199	77.3435015
APR.Severity.of.Illness.Code4	0.68821357	0.0277244	24.8233887
APR.Severity.of.Illness.Code3	0.41723969	0.01895092	22.0168599
APR.MDC.DescriptionDISEASES AND DISORDERS OF THE MUSCULOSKELETAL SYSTEM AND CONN TISSUE	0.45450362	0.04554303	9.97965318
APR.Severity.of.Illness.Code2	0.13866384	0.01692735	8.19170526

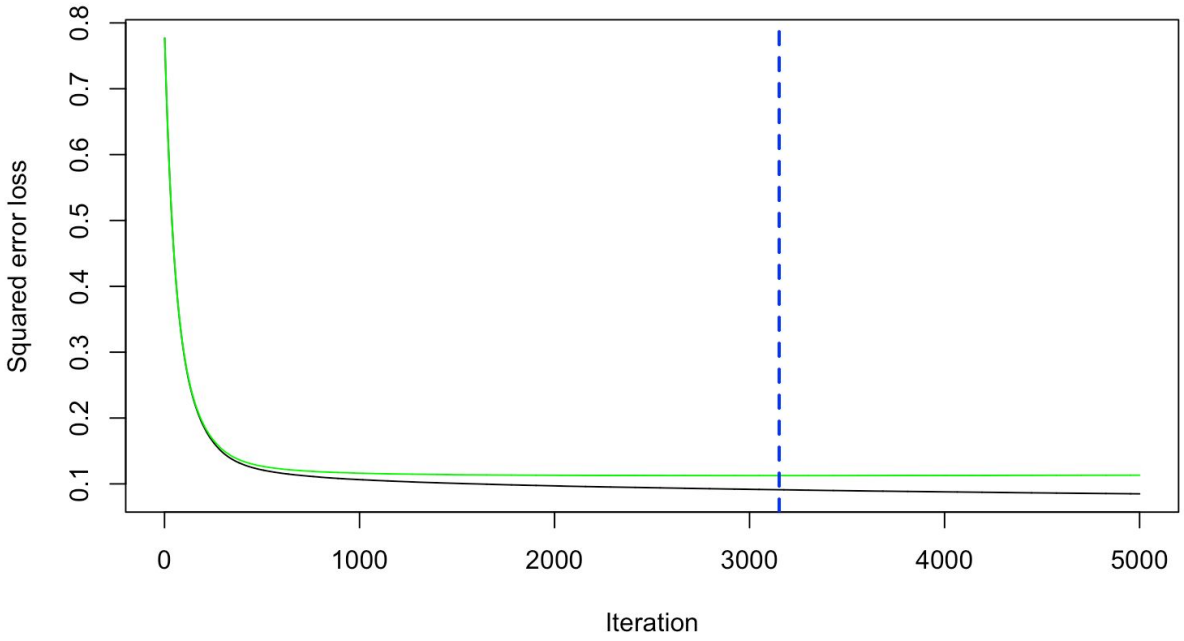
Predicted vs Actual Plot

- A well fitting model should hug the diagonal.
- This one is not really
- Will compare models at end
- Weak correlation => uncaptured variation



Gradient Boost

Ensemble tree

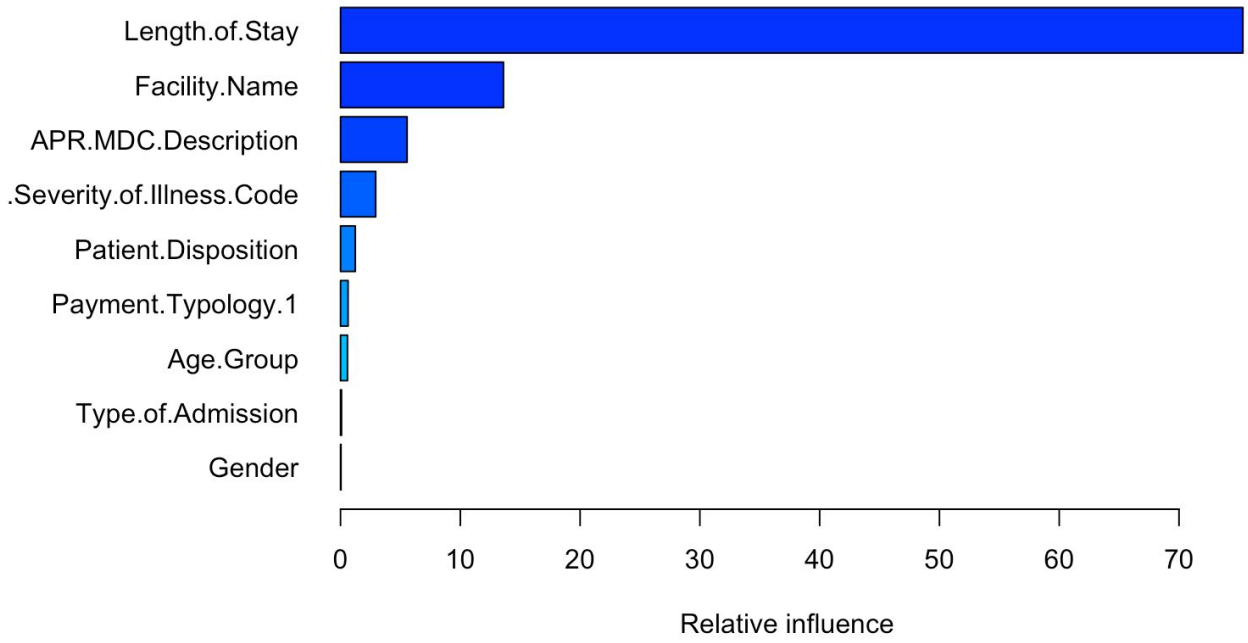


Fitting

- Used cross validation to find the optimal number of trees
  - => 3152
- Using interaction depth 4, shrinkage 0.01

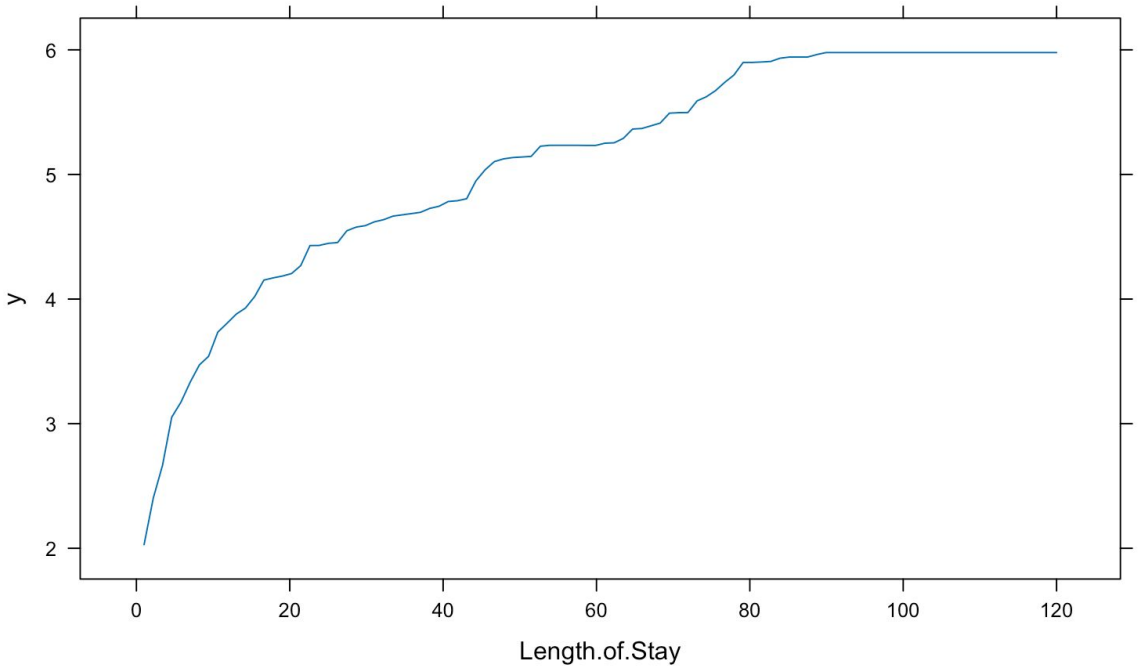
Variable Importance

- Importance shows the variability captured by each predictor



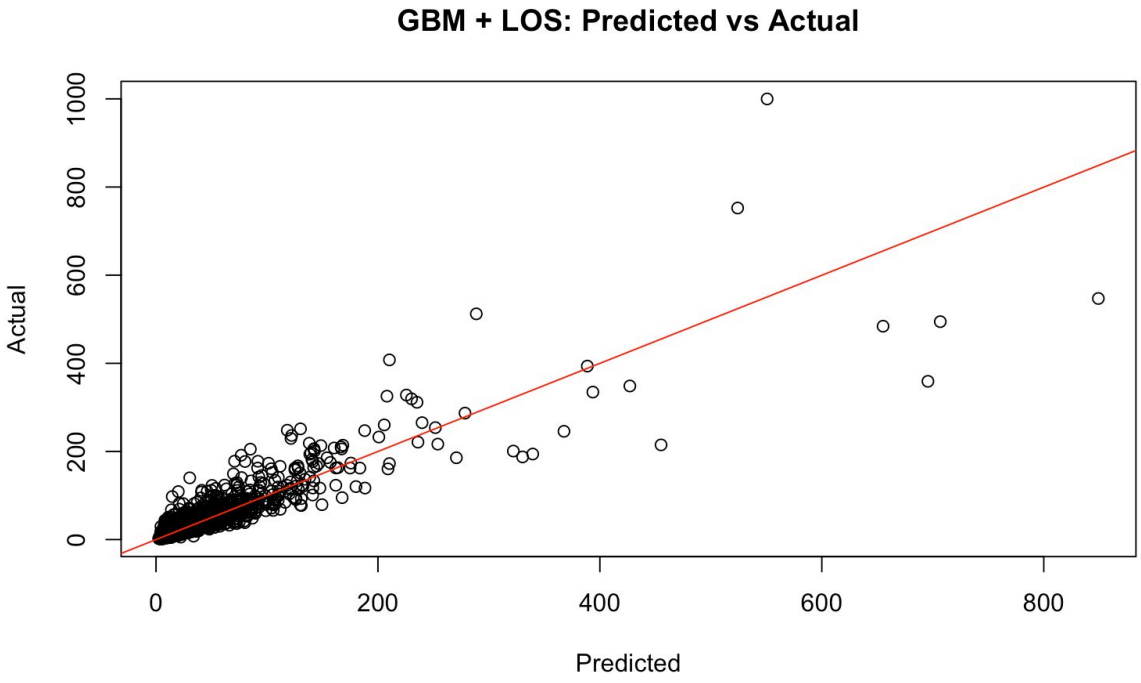
Length of Stay

- Highly correlated with the total cost of the visit



Predicted vs Actual Plot

- A well fitting model should hug the diagonal.
- This one does really well
- Outliers have less data so deviation makes sense



Final Comparison

Model	**MSE**	**R2**
Gradient Boost + Length of Stay	454.8217	0.8008
Gradient Boost	1656.9237	0.3145
Random Effects	1695.1156	0.291
Random Effects + Length of Stay	371609.468	0.2275

Future directions

- Random slope on length of stay for each hospital
- Log transforming the length of stay as well